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AI Impact on Health Equity for Marginalized, Racial, and Ethnic Minorities

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ABSTRACT

Predictive analytics technologies like machine learning, AI and Generative AI models like Large Language Models (LLMs), have garnered enthusiasm for their potential to improve healthcare services in smart cities. However, these rapidly developing intelligent agents that guides the healthcare decisions may also risk exacerbating health inequities along racial, ethnic, gender, and socioeconomic lines, reflecting systemic discrimination ingrained within healthcare practices. Flawed or injudiciously applied AI systems could improperly restrict opportunities and provide substandard care for minority groups by propagating historical patterns of prejudice encoded within limited training datasets. These advanced intelligent technologies can hinder the sustainable health solutions for smart cities. This study examines intelligent AI models and applications in healthcare settings, with a focus on assessing impacts on marginalized and disadvantaged populations. Comprehensive scholarly database searches identified 45 relevant studies investigating issues on algorithmic bias, lack of diverse training data, and discrimination risks linked to healthcare AI systems. The review finds most applications still lack adequate safeguards to prevent discrimination against vulnerable populations. Through the review of these systems, we propose an integrated inclusive smart health model that considers both technical interventions as well as broader participatory and ethical approaches. Realizing AI's fullest potential to meaningfully advance health justice requires not only algorithmic adjustments to mitigate bias, and efforts to improve diversity of training data, transparent analysis frameworks, and best practices for ensuring just AI systems in healthcare, but also a human-centered commitment to thoughtful, inclusive development approaches that center the needs and priorities of communities impacted by health disparities from the outset.

CCS CONCEPTS

• **Social and professional topics → Governmental regulations**; **User characteristics**; **Medical information policy**; • **Computing methodologies → Artificial intelligence**.

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KEYWORDS

AI and healthcare, Health Equity, Algorithmic Bias, Machine Learning in health decision support, Trustworthy AI

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1 INTRODUCTION

Health inequities refer to unfair, avoidable differences in health outcomes and access to care across social groups. As [45] underscore, health disparities reflect structural injustices shaped by racism, bias, and social determinants, not just random variation. For instance, diabetes prevalence is 77% higher among Indigenous groups than the general US and Canadian populations [35]. Overall mortality rates for conditions like heart disease, diabetes, and cancer are markedly higher among racial/ethnic minorities compared to white populations [42]. These deep-rooted health inequities are unacceptable and demand policy changes. Some believe emerging AI technologies like machine learning models could help tackle disparities by improving access, personalizing care, and augmenting clinical decision making [9, 27]. Early AI applications show initial promise for accelerating tasks such as cancer screening, sepsis prediction, and mental health treatment matching [41]. But as an evolving technology, AI also poses acute ethical risks involving discrimination, fairness, accountability, and transparency that merit further discussion [21, 30].

Most troubling is the prospect that flawed AI systems could perpetuate marginalization of disadvantaged groups. If developers fail to address systemic biases, algorithms can easily amplify discrimination [48]. Since AI learns from data, models will absorb any prejudices ingrained within training datasets [3, 39]. Medical data often reflects historical discrimination through underrepresentation of minorities in research and potentially biased disease labels [23, 40]. Without thoughtful design, AI risks automating exclusion of marginalized patients.

In assessing AI and health equity, a balanced perspective is essential. Despite its transformative potential, AI alone cannot resolve systemic biases and discrimination [11]. In fact, AI systems that fail to account for social determinants could misattribute health disparities solely to individual behaviors or genetics, obscuring the external barriers and structural racism facing marginalized communities [16]. Thoughtful input from diverse stakeholders is imperative to ensure AI promotes health equity rather than reinforcing the status quo.

This paper examines the intersection of AI, systemic bias, and persistent health disparities impacting communities of color, low-income groups, gender and sexual minorities, people with disabilities, and other disadvantaged demographics. Using a literature survey method, this paper highlights technical and ethical dimensions of applying AI to assess risks that AI may also exacerbate discrimination against vulnerable populations. It intends to propose how to harness AI to dismantle injustice, while avoiding pitfalls that could perpetuate discrimination under the influx of technology. Achieving health equity with Artificial intelligence demands technical mitigations as well as diverse community engagement, collaboration, and social/cultural contextual understanding, inclusive governance centered on patients from disadvantaged communities. An additional goal is to help to set the AI research agenda for improving health equity.

2 RELATED WORK

This section briefly overviews key studies on sources of bias, their impact on marginalized groups, and emerging approaches to improve fairness, accountability, and inclusivity. A growing body of literature reveals how algorithmic bias and lack of diverse data undermine performance of AI systems and exacerbate discrimination in healthcare.

Most AI systems rely on detection of statistical patterns and correlations in data. As a result, they inherit any biases embedded within their training data [28]. Unfortunately, healthcare data reflects longstanding discrimination and exclusion of marginalized groups. Under-representation in health research and barriers to accessing care means minorities appear less frequently in datasets [4, 40]. Additionally, medical labeling practices and knowledge often codify racial, gender, and other biases [31]. When AI learns relationships from such data, it absorbs and amplifies these biases [9].

Several alarming examples demonstrate how unconscious biases creep into healthcare algorithms. In a widely cited case, [31] found an algorithm for predicting patient health risks exhibited significant racial bias, assigning higher risk scores to Black patients compared to white patients with identical clinical profiles. Studies have also uncovered biased AI tools for skin cancer screening, cardiovascular risk prediction, kidney function tests, and other applications that disadvantage women, racial minorities, and older adults [2, 14, 17] . If deployed uncritically, such biased algorithms could improperly restrict treatment access or provide substandard care to marginalized groups. Some researchers are investigating techniques to promote fairness and equity in AI systems, such as adjusting algorithmic formulas to offset discriminatory effects [36].

Beyond technical factors, leveraging AI's potential to improve health equity requires examining broader ethical implications. Applications of AI must actively seek to reduce disparities and promote justice, not simply avoid direct discrimination [13, 25]. But conflicting priorities around privacy, accountability, and profits may pose obstacles to equity-focused design. Moreover, affected communities currently have little voice in AI decision-making [26], while deployment of AI in resource-poor settings often neglects local needs and priorities [43]. Community-engaged design processes that involve impacted groups in developing and evaluating AI tools are proposed to enhance relevance for underserved populations [18]. Still, comprehensive standards and governance frameworks for equitable AI remain underdeveloped [33].

3 METHODOLOGY

This literature review followed a systematic methodology to identify, evaluate, and synthesize relevant studies on the intersection of emerging technologies, health equity, and marginalized groups (if this is done by others, you need to cite it. Here). Six scholarly databases (PubMed, EMBASE, Web of Science, Google scholar, ACM Digital Library, and CSI library) were comprehensively searched to find pertinent studies.

Search terms included keywords and controlled vocabulary related to three concepts: 1) emerging technologies, algorithms, automation; 2) health equity, disparities, bias, discrimination; and 3) marginalized populations. The search was limited to English language studies published between 2010-2023. Reference lists of eligible studies were hand-reviewed for additional relevant sources. Studies were included if they examined applications of emerging technologies in healthcare and discussed impacts on equity, disparities, bias, or discrimination affecting marginalized populations. Exclusion criteria were non-peer reviewed articles, technologies unrelated to healthcare, and lack of focus on equity impacts. Table 1 summarizes the data collection with search criteria, and the inclusion and exclusion criteria.

Titles and abstracts of retrieved studies were independently screened by two reviewers based on the eligibility criteria. After removing duplicates and irrelevant sources, full text articles were assessed to confirm inclusion. Overall, 45 studies met the eligibility and quality criteria to be included in this literature review. They encompassed diverse methodologies including audits, case studies, interviews, literature analyses, and technical papers (as shown in Figure 1.

For further analysis, we used a standardized form to extract key data from included studies, such as study design, technology application area, sample characteristics, main analyses/metrics, findings related to equity impacts, limitations, and conclusions. A narrative synthesis approach was used to analyze the extracted data. Major themes related to technologies and health equity were summarized and compared across studies. Tables and graphical plots were generated to visually depict findings.

The selected articles are represented by type, and by time distribution as shown in Figures 1 and 2.

4 FINDINGS

4.1 Challenges in Health Equity

4.1.1 Algorithmic Biases in Health Systems. Table 2 summarizes the selected studies that demonstrate how algorithmic bias has been manifested in thematic areas of healthcare applications. Several key datasets have been utilized in these studies to quantify and characterize algorithmic biases in healthcare AI systems.

A widely cited analysis by [31] found a commercial algorithm to predict healthcare costs exhibited significant racial bias. Controlling for clinical factors, the model assigned Black patients risk scores averaging 46% higher than equally sick white patients. This demonstrates how biases become baked into AI systems. [2] reviewed

Table 1: Summary of data collection methods

Databases searched	PubMed, IEEE Xplore, ACM Digital Library, Google Scholar, CSI library
Search terms	("artificial intelligence" OR "machine learning" OR "neural network*" OR algorithm) AND ("health equity" OR disparity OR bias OR discriminate) AND ("marginalized" OR "minority" OR "vulnerable" OR "underserved")
Date limits	January 2010 - November 2023
Inclusion criteria	Peer-reviewed studies on AI/ML applications in healthcare that examined impacts on equity, disparities, fairness, or related issues.
Exclusion criteria	Non-peer reviewed articles, non-healthcare AI, lacked focus on equity impacts.

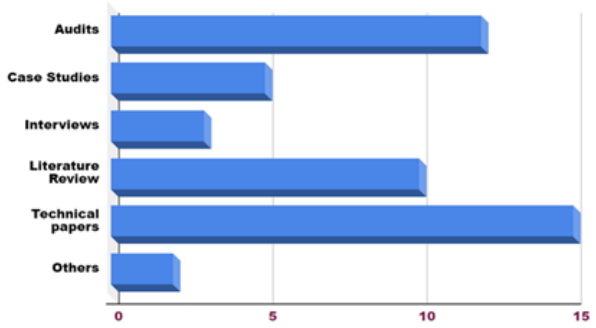

Figure 1: Selected articles by type

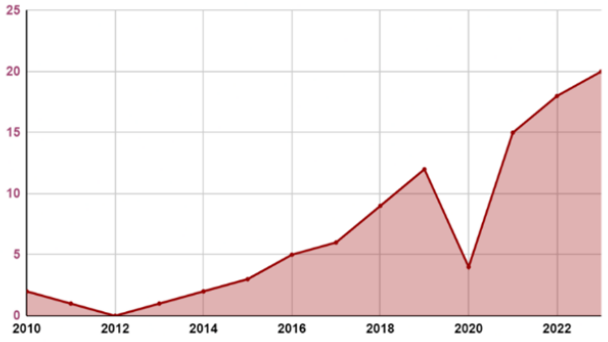

Figure 2: Selected Articles by publication counts by year

machine learning studies for skin cancer screening, finding most models performed worse for minorities, older adults, and females. For example, algorithms classified benign moles as malignant at higher rates for Black patients versus white patients.

An analysis of AI knee osteoarthritis assessment tools found worse performance for female presenting symptoms [14]. Tools were 5% less accurate for women, demonstrating gender bias. Cardiovascular risk algorithms exhibit age and gender bias, overestimating risk for older adults and underestimating risk for younger women [17].

Table 2 also shows key example datasets used in the algorithms, the types of algorithms trained and tested on them, and statistics illustrating the racial, gender, or other biases identified against marginalized groups. Data sources range from ICU patient data to large genomic datasets. Analyses of these datasets consistently reveal poorer performance of algorithms for minorities, women, and vulnerable populations compared to advantaged groups. For instance, an ICU deterioration warning system had 14% worse accuracy for black patients; and a kidney function test was shown to misestimate function for blacks by up to 21%. The sample statistics highlight the varied forms of bias that emerge from unrepresentative training data, prejudiced labeling practices, and imbalance across classes.

Generative AI models such as Large Language Models (LLMs)– ChatGPT (OpenAI) or Gemini (Google) – have also shown risks of negatively affecting health inequities. Despite accurately answering many medical questions [24, 38], analysis shows ChatGPT retains problematic biases potentially improperly influencing care for marginalized groups [8]. For example, [29] gave ChatGPT patient vignettes with identical concerning symptoms but varying demographics. ChatGPT correctly triaged insured patients with high-risk chest pain to emergency care. However, for uninsured patients, it split recommendations between emergency departments and community clinics, although symptoms warranted immediate emergency evaluation [29].

Numerous research characterizes various biases propagated through ChatGPT and LLMs, including preferential behaviors benefiting certain gender, racial, ethnic, and age groups [5, 7]. For instance, some skin cancer algorithms perform worse for minorities and older adults [14]. Content analysis also reveals LLMs can disproportionately amplify stereotypes and ideological stances [6, 15].

These examples underscore risks as models statistically learn relationships from training data. Any distortions or underrepresentation encode prejudices for chatbots to inherit [3]. Simply omitting explicit protected attributes like race or gender from training data does not solve algorithmic bias issues because subtle proxies in the data, like zip codes, can perpetuate prejudices and enable unfairness to endure [23]. Lasting solutions require embracing collaborative frameworks elevating impacted voices [18, 21] and cautious implementation of chatbots complementing providers' essential discretion (Harrer, 2023).

4.1.2 Lack of Diverse Data. In addition to absorbing biases, AI systems suffer from lack of diverse and inclusive data. Minority groups have historically been underrepresented in health research

Table 2: Example Studies on Algorithmic Biases

Dataset	Description	Algorithm Task	Bias Identified	Source
PhysioNet ICU Data	Clinical ICU data	Patient deterioration warning	Algorithm had 14.1% worse AUROC for Black patients	[20]
UK Biobank	500,000 participants' health data	Disease risk prediction	South Asians had 12% higher CAD risk scores than white British	[34]
MIMIC ICU Database	40,000 ICU stays data	Kidney function test	Algorithm misestimated GFR for Black patients by up to 21%	[42]
ISIC Archive	20,000+ dermatology images	Skin cancer screening	9.3% higher error rate for females vs males	[2]
eICU Database	200,000+ ICU admissions data	Sepsis prediction	AUC 5% lower for females than males	[1]
Clinical Practice Research Datalink	Primary care EHR from UK	Population analysis	95% missing ethnicity data	[32]
EHR and claims data	Patient cardiovascular data	Cardiovascular disease risk prediction	Overestimated risk for older adults, underestimated risk for younger women	[17]
Knee imaging and symptom data	MRIs and patient symptoms	Knee osteoarthritis assessment	5% lower accuracy for female patients versus male patients	[14]
Dermatology image datasets	Images of skin lesions	Skin cancer screening	Higher false positive rates for Black patients than white patients	[2]
Commercial insurance claims data	Health records of patients	Healthcare cost prediction	Black patients assigned 46% higher risk scores than similar white patients	[31]

and data collection [4]. One study found non-white ethnicities were completely absent from over 90% of UK medical datasets and race/ethnicity data is also frequently excluded or inconsistently recorded in electronic health records [37].

This presents dual problems for equitable AI. First, it limits model understanding of how symptoms and diseases manifest differently across populations. Without diverse data, AI risks missing unique health patterns or needs for marginalized groups. Secondly, exclusion from data means AI models are less tested and validated for minorities. Performance disparities highlighted earlier result from lack of adequate minority representation.

4.2 Approaches to Improve Health Equity

Given challenges with algorithmic bias and data limitations, research studies explore both technical and other strategies to develop equitable AI systems, including bias mitigation methods, improved data collection, and participatory design with marginalized communities.

4.2.1 Algorithm Adjustment Techniques. Adjusting the statistical formulas and training process allows correcting biases identified in AI models:

- Optimized input data pre-processing can improve model convergence for minority groups and enhance fairness [36] by methods such as targeted data augmentation and reweighting to reduce disparities or imbalance before training algorithms [28]. These methods include, oversampling underrepresented groups in the training data helps minimize class imbalance which involves duplicating minority samples to boost their signal in the data; SMOTE (Synthetic Minority Oversampling Technique) that algorithmically generates additional synthetic samples from minority groups to improve representation; 3) Re-weighting that assigns higher importance weights to minority groups during model training to counteract imbalance. These data sampling and weighting methods help prevent the algorithm from learning biases against excluded groups or overfitting to the majority class.
- During in-processing stage, algorithms can be re-trained after bias quantification until performance is equalized across groups [28]. Emerging in-processing techniques apply fairness constraints and adjustments during the model training process itself to promote equity [36]. These include: 1) Parity constraints to optimize model parameters until performance equality is achieved across groups. For instance, constraining skin cancer classification accuracy to be equal for white and Black patients; 2) Regularization terms to add fairness penalties to the objective function optimized during training. For example, directly penalizing racial, gender or age discrimination; 3) Adversarial techniques to use competing neural networks to minimize bias "blind spots" in the model. By intervening during the training process, in-processing

techniques can reduce the amount of bias learned by the model compared to unconstrained training.
- Post-processing model outputs using recalibration techniques can mitigate biases. These methods include: 1) Threshold optimization that adjusts the decision threshold for model outputs to equalize performance. For example, changing score cutoffs by group to ensure equitable screening; 2) Calibration techniques nudge model outputs towards fairness. This corrects scores from biased models to unbiased predictions [22]; 3) Bolstering that takes a fairness-enhancing model and uses its outputs to correct a biased model. By intervening after initial model training, post-processing approaches offer flexibility to fix biases without needing to re-train models from scratch.

These techniques show promise to enhance algorithmic fairness but require recognizing and quantifying biases first through audits. They also focus narrowly on the model rather than broader data and system issues.

4.2.2 Improved Data Collection and Sampling. In addition to algorithm adjustments, generating more representative training data itself is critical for equitable AI design. Strategies reported in the literature include:

- Targeted sampling and recruitment of diverse participant cohorts for health studies to reduce exclusion of marginalized groups from data [4]
- Enhanced recording of race, ethnicity, gender identity, and other demographic variables in electronic health records to support disaggregation in AI systems [37]
- Collecting complementary data like social determinants through community partnerships to better contextualize disparities [10]
- Synthetic data generation techniques such as generative adversarial networks to augment underrepresented minorities when real-world data remains scarce [47]

While balancing privacy, thoughtful approaches to expand health data diversity such as above are offered to improve algorithmic performance and avoid blind spots for marginalized groups. But equitable data alone is insufficient without fairness-aware algorithms.

4.2.3 Participatory Techniques. Beyond purely technical measures above, researchers advocate participatory design processes that center marginalized communities in developing and governing AI systems to promote equity and justice [18, 21] with the following strategies:

- Inclusive design teams draw on diverse stakeholders and disciplines, not just computer scientists.
- Co-creation practices from impacted communities help formulate problems and evaluate solutions.
- Community oversight boards that center disadvantaged populations in AI governance decisions
- Proactive community engagement from the problem formulation through evaluation.

Engaging marginalized populations directly in co-development and evaluation of AI systems can help ensure their needs are met [46]. Drawing on diverse stakeholders' expertise through forums like participatory workshops allows surfacing ethical issues early [19]. Centering voices of disadvantaged groups in governance of AI systems helps align technology with community values and priorities [21].

When marginalized groups participate meaningfully in shaping AI systems, it becomes more likely the technology will address rather than exacerbate equity problems. However, truly participatory approaches require building trust, addressing power differentials, and overcoming engagement barriers with communities. If done respectfully, they hold promise to align AI with the priorities of excluded populations. Efforts need to be made for actively recruiting and compensating diverse partners. There are also power dynamics requiring sensitive facilitation. If done well, participatory processes could enhance relevance and acceptability of AI for marginalized groups.

While promising, techniques to address algorithmic bias remain underdeveloped and not widely implemented. Most current AI lacks concrete safeguards against discrimination [9]. But growing research recognizes the need for proactive, multi-pronged strategies. Integrating community partnership, diverse data, bias quantification, and technical mitigation offers a potential pathway towards equitable AI design.

5 DISCUSSIONS

From the literature survey, we have found the AI's potential for promoting health equity and risks in perpetuating the equity in healthcare that are summarized in Table 3.

These benefits remain largely theoretical. Most current AI applications fail to proactively incorporate equity considerations [17]. The field requires purposeful design choices, diverse teams, and community guidance to actualize AI's potential to reduce disparities—rather than naively assuming technologies designed for majority groups will automatically benefit minorities. Conversely, studies reveal AI risks exacerbating discrimination through multiple mechanisms if deployed injudiciously. Proactive steps are imperative to ensure AI promotes rather than inhibits health justice.

As Topol argues [41], warranted optimism about AI's potential must be tempered with skepticism until more evidence demonstrates real impact on equity outcomes. Avoiding "techno-solutionism" will be critical. AI alone cannot resolve complex socio-historical drivers of health disparities but could complement broader efforts centered on social justice [3].

Moving forward, the healthcare community should maintain cautious optimism about AI's role in equity. Thought leaders stress the need to continually evaluate benefits against potential for harm throughout development and deployment [9]. Keeping communities at the center will be vital. Recent works underscore that realizing AI's benefits demands inclusive processes rooted in the priorities of marginalized groups [18, 21].

Unjust algorithms often arise when advantaged developers lacking lived experience of disadvantage design systems without diverse voices at the table. In contrast, participatory approaches like cooperative prototyping, user testing, and democratized governance could produce more just AI systems aligned with marginalized communities' self-defined needs [46].

Table 3: AI's Potentials and Risks for Health Equity

Potentials of AI for Health Equity	Risks of AI to Perpetuating Inequity
• Expand access to care through automating tasks and reaching underserved areas [41, 43] • Enable personalized medicine tailored to individuals' uniqueness including genomics, behaviors, and environment [10] • Augment clinical decisions by detecting patterns human providers may miss, avoiding cognitive biases [41] • Analyze diverse health data to identify determinants driving disparities and target interventions [12] • Promote system equity through optimized resource allocation, ensuring fair distribution across groups [11]	• Algorithmic bias results from learning stereotypes and prejudices ingrained in training data [3, 48] • Lack of diverse data causes blind spots in model performance for marginalized groups [9, 40] • Focus on correlation risks confusing causal myths. For example, wrongly attributing health behaviors solely to individual choices while ignoring structural barriers faced by disadvantaged communities [17] • Automating without ethics may scale up existing disparities and remove human judgment [21] • Uncritical trust in AI's objectivity obscures how technologies inherit human biases [30, 44]

However, truly participatory design remains rare, given power differentials, profit motives, and lack of incentives for tech companies. In addition, developing inclusive AI will require re-thinking funding, evaluation, and research practices to center diverse stakeholders, especially groups historically excluded or harmed by past technologies [3]. Table 4 summarizes proposed principles from the literature to guide AI designs.

Scholars argue focusing solely on technical interventions like reducing algorithmic bias risks losing broader contextual understanding of the societal origins of health disparities [23]. For instance, "debiasing" AI models could remove overt discrimination in automated decisions. But this narrowly defined fairness fails to resolve underlying disparities shaped by longstanding oppression.

As Benjamin cautions[3], "the 'bias' is in society, not just machine learning systems." Truly addressing health inequities requires examining how discrimination becomes embedded within data, institutions, culture, and systems - not just individual technologies [42]. AI designers must recognize how technology intersects with this complex historical and social context [17]. Key concerns and possible solutions requiring ongoing research we identified in the literature include:

- Quantifying and mitigating subtle biases that persist even in AI systems adjusted to promote fairness [28];
- Developing more robust, representative health datasets ethically and inclusively [4];
- Ensuring participatory design processes engage marginalized communities meaningfully while addressing power imbalances [46];
- Building governance structures incentivizing equity-promoting AI despite competing interests of powerful technology companies [3]; and
- Evolving regulatory and accountability mechanisms to detect and prevent algorithmic harm [30].

In applying AI to expand access and efficiency, difficult questions arise regarding the ethics of automation [21]. Replacing human roles through automation to cut costs disproportionately impacts lower-resourced communities already facing inadequate care. Irresponsible automation could also scale up discrimination by removing opportunities for nuanced judgment [30]. Relatedly, automation concentrates knowledge and power in technical systems. As AI governs more decisions, marginalized communities may lose autonomy unless granted meaningful say over these technologies [3]. Democratizing the power behind AI will be as vital as enhancing technical fairness.

The research agenda for the health equity by AI posed by various studies include:

- How can marginalized communities gain greater voice in health AI governance? [18].
- What programmatic initiatives or incentives could cultivate more just and inclusive AI systems? [36].
- How do social contexts and power dynamics influence participatory design in practice? [46].
- What regulatory frameworks would compel addressing algorithmic discrimination and equity? [30].
- How can we enhance understanding of community priorities and values to guide health AI? [21].
- What interdisciplinary partnerships and approaches are required to progress responsibly? [3].

6 CONCLUSIONS

This paper examined recent publications from 2010 to 2023 on AI and Health equity to spur deep dialogue on AI's complex role in promoting health equity for marginalized groups. We presented the literature selection process and findings. The lessons learned include the health equity by AI stems from various sources, such as algorithmic biases in collecting, sampling, and training models with data lacking diversity of marginalized groups. The biases in data are percolated to the models that can perpetuate the cycle.

Many remedies of technical and participatory engagement are also investigated from the literature with data balancing, fairness

Table 4: Principles for Equitable Health AI

Principle	Description	References
Stakeholder participation	Meaningfully engage impacted communities in co-design and evaluation	[18]
Transparency	Openly share model development processes and performance data	[9]
Holistic assessment	Consider social contexts and impacts beyond technical metrics	[21]
Regulations and oversight	Establish governance frameworks to align AI with health equity	[3]

constraint checks, as well as data collections through participation of underrepresented groups. The Generative AI, a new super-intelligent tool like ChatGPT, is no exception to the inequities in health research and practices.

This literature survey suggests the AI design, modeling, and assessment principles, such as inclusive design, co-creation, transparency requirements, contextualized understanding of the community, and governance model to regulate AI for health equity.

The future work for enhancing the AI's health equity is also suggested: Empowering marginalized communities to gain greater voice in health AI governance; Initiating Programs or incentives to cultivate more just and inclusive AI systems; Leveraging social contexts and power dynamics to influence participatory design in practice; Proposing regulatory frameworks to address algorithmic discrimination and equity; Enhance understanding of community priorities and values to guide health AI; Forging interdisciplinary partnerships and approaches to progress responsibly. Based on this preliminary study, our theoretical framework for inclusive equitable health AI framework is planned as our future work.

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